

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: E.G.G (Employment Guidance Generator)

1.0 Project Overview

- **Problem Statement:** Students often make career and education decisions based on incomplete information, social pressure, or generic advice, leading to poor financial outcomes such as unemployment, student debt, or low-return degrees. Existing tools provide static recommendations (e.g., personality quizzes) but lack real-time, context-aware insights that combine labor market trends, personal strengths, and financial implications.
- **Target Domain:** AI for Economic Empowerment & Decision Intelligence, in education and career planning.
- **Proposed Solution Summary:** Omelette Squad's E.G.G leverages Z.AI's GLM to analyze both structured (salary data, job demand, tuition costs, market statistics) and unstructured inputs (student interests, goals, personal constraints) to deliver personalized, explainable career pathways. It generates actionable recommendations, forecasts financial outcomes, and provides decision trade-offs—enabling students to make informed choices that optimize long-term economic success.

2.0 Background & Business Objective

Background of the Problem: Career decision-making today is fragmented. Students rely on scattered sources—career websites, anecdotal advice, outdated statistics—without a unified system to synthesize insights. Many lack access to career counselors, and existing digital tools fail to incorporate real-world economic signals such as job market demand, salary progression, and cost-benefit analysis of education paths.

Importance of Solving This Issue: Poor career decisions can lead to long-term financial instability. By equipping students with AI-driven insights, we can:

- Reduce mismatched career choices
- Improve employability outcomes
- Minimize unnecessary education costs
- Increase lifetime earning potential

Strategic Fit / Impact: Aligns with Z.AI's vision of embedding GLM into real-world decision-making systems. E.G.G demonstrates how GLM transforms raw data into **decision intelligence**, directly impacting economic mobility and workforce optimization.

Competitive Differentiation:

Unlike traditional career quizzes or static tools (e.g., personality-based assessments or generic career portals), E.G.G integrates:

- Real-time labor market data
- Financial outcome simulation (ROI, break-even)
- AI-driven reasoning with explainable trade-offs

This positions E.G.G as a decision intelligence system rather than a recommendation tool.

3.0 Product Purpose

3.1. Main Goal of the System:

To provide an AI-powered decision intelligence assistant that helps students choose optimal career and education paths based on personalized data and economic outcomes.

3.2. Intended Users (Target Audience):

- High school students
- University students
- Career switchers (early-stage professionals)
- Parents who are guiding their children
- Academic advisors / counselors

3.3 Success Metrics

The effectiveness of E.G.G will be evaluated using the following measurable indicators:

- Recommendation Relevance Rate: $\geq 80\%$ of users agree that suggested careers match their interests and strengths.
- Economic Optimization Score: Average projected ROI improvement compared to baseline (non-guided decisions).
- User Completion Rate: $\geq 90\%$ of users successfully complete the full input $\rightarrow$ recommendation flow.
- Decision Confidence Increase: Users report higher confidence in career decisions after using the system.

4.0 System Functionalities

4.1. Description:

The system acts as a decision intelligence engine. It ingests user inputs (interests, grades, financial constraints) and combines them with external datasets (job market trends, salary benchmarks, education costs). Z.AI GLM interprets this data, performs reasoning, and generates insights, forecasts, and recommendations.

Usage Flow: Input → AI Analysis → Career Recommendations → Comparison → Decision Support

Users first provide their profile and optional CV, after which the system generates ranked career options. Users can then evaluate trade-offs and financial outcomes before making informed decisions.

4.2. Key Functionalities:

Career Path Recommendation Engine: Suggests optimal career options based on user profile and market demand.

Financial Outcome Simulator: Estimates ROI of different education/career paths (e.g., tuition vs expected salary over time).

Scenario Comparison: Allows users to compare multiple career paths (e.g., Computer Science vs Business) with trade-offs.

Personalized Skill Gap Analysis: Identifies skills needed and recommends learning pathways.

Explainable Insights: Provides clear reasoning behind recommendations (why a career is suitable or risky).

4.3. AI Model & Prompt Design

4.3.1. Model Selection:

Z.AI GLM is used due to its strengths in:

- Context-aware reasoning across structured and unstructured data
- Natural language explanation of complex decisions
- Multi-step inference for trade-off analysis

- Ability to generate actionable insights rather than raw outputs

Without GLM, the system reduces to static dashboards without meaningful decision support.

4.3.2. Prompting Strategy:

- **Multi-step agentic prompting**
- Pipeline approach:
 1. Input interpretation (student profile extraction)
 2. Data synthesis (market + personal context)
 3. Reasoning (trade-offs, forecasting)
 4. Recommendation generation
 5. CV upload (CV will be analysed by Z.AI)

This ensures deeper decision intelligence rather than shallow responses.

4.3.3. Context & Input Handling:

- Accepts structured inputs (forms) + unstructured inputs (free text goals)
- Large inputs are **chunked and summarized** before processing
- Maximum input handled: ~8k–16k tokens (depending on deployment)
- Excess input is:
 - Summarized using GLM
 - Prioritized by relevance scoring

4.3.4. Fallback & Failure Behavior:

- Confidence scoring for recommendations
- If hallucination risk detected:
 - System retries with constrained prompts
 - Falls back to data-driven templates
- Provides disclaimer when uncertainty is high
- Optional human advisor escalation

5.0 User Stories & Use Cases

User Stories:

- "As a student, I want to input my interests and strengths so that I receive personalized career recommendations."
- "As a student, I want to compare career options so I can choose the most financially viable path."
- "As a parent, I want to understand the ROI of my child's education choices."

Use Cases (Main Interactions):

Career Recommendation Flow:

User inputs profile → GLM analyzes → System generates ranked career options with explanations

Financial Simulation Flow:

User selects career → System forecasts salary trajectory, education cost, ROI

Comparison Flow:

User selects multiple careers → GLM generates trade-off analysis

6.0 Features Included (Scope Definition)

- AI-powered career recommendation engine
- CV analyser
- Financial outcome simulation dashboard
- Scenario comparison tool
- Explainable AI insights (natural language explanations)
- Integration with job market datasets

7.0 Features Not Included (Scope Control)

- Direct job placement or recruitment services
- Real-time university admission processing
- Fully automated decision-making (human still decides)
- Multi-user collaborative planning (initial version)

8.0 Assumptions & Constraints

LLM Cost Constraint:

- Estimated cost per session: medium (multi-step reasoning required)
- Optimization strategies:
 - Caching repeated queries (e.g., common careers)
 - Summarizing inputs before full processing

Technical Constraints:

- Accuracy depends on external data quality (requires constant datasheet update)
- Limited real-time updates for job market data

Performance Constraints:

- Complex simulations may take a few seconds
- Token-heavy prompts limit concurrency

User Input:

- Requires honest and accurate input from users
- Not a replacement for professional counseling

9.0 Risks & Questions Throughout Development

Bias in Recommendations: How to ensure fairness across socioeconomic backgrounds?

Data Accuracy: What happens if job market data is outdated?

Over-Reliance on AI: How to prevent users from blindly following recommendations?

Explainability vs Complexity: Can insights remain simple while handling complex trade-offs?